Horizontal launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find flight time from vertical motion and then horizontal range.

SUCCESS CRITERIA - I can...

1. Separate vertical and horizontal givens.
2. Find or use fall time before calculating range.
3. Check units and interpret the landing point.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A spherical planet has surface field 75 N/kg. Find the field at center distance 5 R, where R is the planet radius.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $h = \frac{1}{2}gt^2$; range = $v_x t$.
Horizontal launch; h is the positive drop, t is fall time; g = 10 m/s². CPA may use the supplied time.

Worked reasoning: A horizontal launch falls 20 m with g=10 m/s². From $20=5\,t^2$, t=2 s. At horizontal speed 3 m/s, range=6 m.

Your turn: Draw a diagram labeling height and range. Number the two solution steps to show why time comes first.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and interpret the model

On the diagram, label height h, horizontal range and initial horizontal velocity. Which direction determines the fall time?



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

In a launch simulation, two balls leave the same platform horizontally. Both fall for 2 s. One launches at 4 m/s and the other at 6 m/s. Draw and label both paths; calculate each horizontal range.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

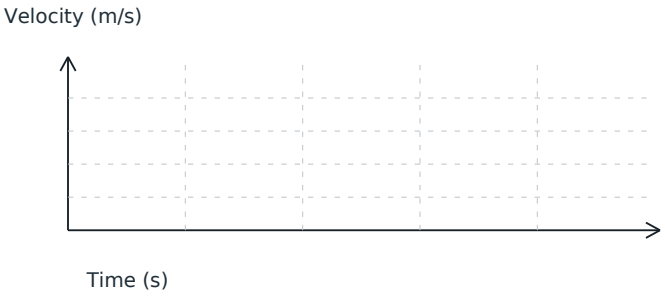
05 Explain your reasoning

Two platforms have heights 5 and 20 m. Supplied fall times are 1 and 2 s. At the same launch speed, does quadrupling height quadruple range? Explain.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 6 kg cart starts from rest under 48 N net force right for 5 s. Find acceleration and final velocity. State one assumption.
Label the given and predicted quantities on the model.



Marzano check - circle: 1 2 3

07 Exit check

A horizontal launch at 5 m/s stays in air 3 s. Where does it land relative to the platform base? Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A target is 10 m from a platform whose fall time is 2 s. Choose the horizontal launch speed, then predict where half that speed lands. Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3